HACKSPHERE 3.0 · ATMOS ’26 · BITS PILANI HYDERABAD

Gridwise

Smart Energy Microgrid and Demand Forecasting Engine

Hourly demand forecasting and power-source transition control, deployed on Zoho Catalyst.

Team Kryptonite

October 2026

Zoho Catalyst · AppSail · QuickML · Data Store

Requirements and implementation

Requirement	Implementation	Status
Regression model trained and published in QuickML	CatBoost and linear regression, averaged; published QuickML endpoint	Complete
Telemetry ingestion by form or CSV, with cleaning	CSV upload, hourly editor, manual form; validation and gap filling	Complete
24-hour demand forecast from the QuickML endpoint	Batched calls from AppSail for 24 consecutive hours	Complete
Threshold and time-based transition controller	Storage mode above 160 kW or 18:00–22:00; alert at 200 kW	Complete
Dashboard: load curve, grid state, battery SOC, alerts	Overview, dispatch schedule, controller log, SOC chart	Complete
Persistence in Data Store (recommended)	ForecastRuns, Telemetry and AlertLog tables	Complete
predict.py for automated evaluation	Calls the deployed model; organisers’ self-test passes	Complete

Evaluation weights: forecast accuracy 35% · deployment 35% · controller logic 20% · dashboard 10%

02 · ARCHITECTURE

System architecture



Component	Catalyst service	Responsibility
Dashboard and REST API	AppSail (FastAPI, Python 3.12)	Input validation, cleaning, orchestration
Demand model	QuickML endpoints gridwise-demand-catboost and gridwise-demand-linear	Hourly demand prediction
Persistence	Data Store: ForecastRuns, Telemetry, AlertLog	Runs, inputs, predictions, controller events
Evaluation interface	predict.py → /api/predict	Model access without credentials
Authentication	Platform credential injected by AppSail	No keys in the browser or repository

Model selection with chronological validation

FOUR EXPANDING PAST-TO-FUTURE FOLDS ON THE FIRST 3,200 HOURS · FINAL 800 HOURS LOCKED UNTIL SELECTION

QuickML algorithm	Configuration	Mean fold RMSE	Locked holdout RMSE
CatBoost regression (selected)	**1,000 iterations · learning rate 0.03 · depth 3**	**9.97**	**11.01**
XGBoost regression	1,200 trees · learning rate 0.03 · depth 3	10.69	13.16
LightGBM regression	800 trees · learning rate 0.03 · 7 leaves	10.82	15.31
Random forest regression	QuickML defaults (100 trees)	10.96	12.57
Starter-kit formula	Provided baseline	–	45.85

QUICKML 5-FOLD CROSS-VALIDATION ON CONSECUTIVE DATE BLOCKS

QuickML model	Mean RMSE	Most recent block	Maximum error
CatBoost regression (deployed)	**9.23**	**10.74**	**54.9**
AutoML (GB regression with Autolearn)	9.83	12.63	61.0

RMSE in kW, lower is better · deployed model holdout R^2 0.968

Forecast accuracy on the most recent data

Each model is refit on the hours before a split and scored on the hours after it. The most recent weeks average 29.1 °C, matching the 29.2 °C of the evaluation sample.

Model	Status	Last 800 h	Last 600 h	Last 400 h
CatBoost regression (QuickML)	Deployed	10.74	9.57	8.64
Linear regression, 52 features	Deployed	9.27	9.51	8.58
Equal-weight average of both models	**Deployed**	**9.22**	**9.03**	**8.40**
Starter-kit formula	Baseline	45.85	–	–
RMSE reduction: average vs CatBoost alone		14.2%	5.6%	2.8%

RMSE in kW · CatBoost holdout RMSE varies from 10.68 to 11.07 kW across random seeds · averaging weight fixed at 50/50

Transition controller rules and verification

Rule	Condition	Output
R1 · Load threshold	Predicted demand above 160 kW	Renewable Storage Mode
R2 · Evening window	Hour from 18:00 to 22:00, inclusive	Renewable Storage Mode
R3 · Default	Neither R1 nor R2 applies	Main Grid Mode
R4 · Overload	Predicted demand of 200 kW or more	Critical Overload Alert, independent of mode

Verification	Cases	Result
Every hour of the day at 0, 120, 159.99, 160, 160.01, 199.99, 200 and 260 kW, checked against a reference implementation of the rules	192	All passed
Complete automated test suite, including 643 property and matrix tests	891	All passed
Mode switches and alerts written to the AlertLog table on the live deployment	6 events	Verified

Source: engine/controller.py · tests/test_core.py · tests/test_properties.py

Controller output on a live forecast

24-hour window beginning 16 June 2015 at 15:00, forecast by the deployed QuickML model on the live Development deployment.

Metric	Value
Peak demand	294.6 kW at 18:00
Minimum demand	91.8 kW at 03:00
Energy over 24 hours	4,730 kWh
Renewable Storage Mode	16 hours
Main Grid Mode	8 hours
Critical Overload Alert	15 hours
Mode transitions	2

Period	Demand, kW	Grid mode	Alert
16 Jun 15:00–17:00	234.5–245.9	Renewable Storage	Yes
16 Jun 18:00–22:00	222.1–294.6	Renewable Storage	Yes
16 Jun 23:00–17 Jun 06:00	91.8–149.4	Main Grid	No
17 Jun 07:00	188.6	Renewable Storage	No
17 Jun 08:00–14:00	208.3–262.9	Renewable Storage	Yes

Source: control schedule exported from the live dashboard (QuickML ensemble: CatBoost + linear regression)

Battery storage scenario analysis

The same forecast re-simulated at six storage capacities, without additional model calls. Ratings: 200 kW discharge, 150 kW PV, 95% efficiency, 20% reserve.

Battery capacity	Energy from battery	Grid support required	Hours needing grid support
300 kWh	206 kWh	3,236 kWh	16 of 16
600 kWh	377 kWh	3,065 kWh	16 of 16
900 kWh	548 kWh	2,894 kWh	16 of 16
1,200 kWh (default)	719 kWh	2,723 kWh	16 of 16
1,800 kWh	1,061 kWh	2,381 kWh	16 of 16
2,400 kWh	1,403 kWh	2,039 kWh	16 of 16
5,977 kWh at 293 kW (recommended)	**Covers all storage hours**	**0 kWh**	**0 of 16**

Source: live /api/scenarios on the forecast shown on the previous slide · capacity found to the nearest kWh

Fault tolerance and safeguards

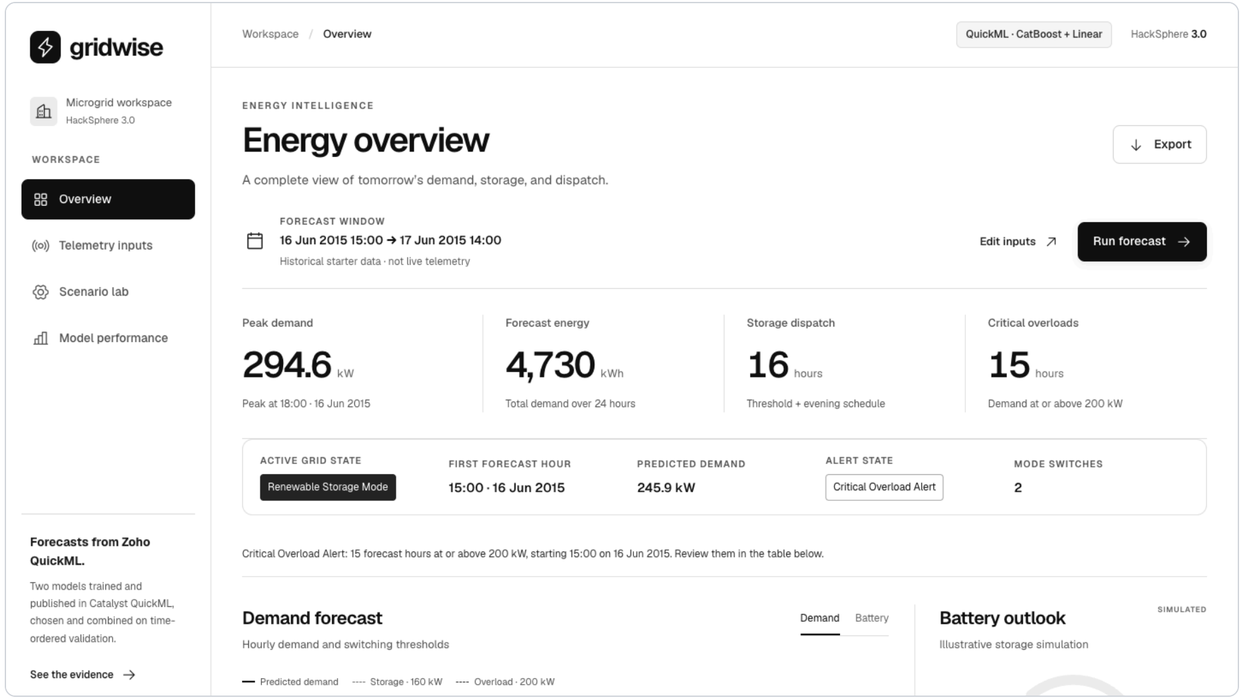
Condition	System behaviour
One QuickML endpoint unavailable	The remaining model answers alone, and the label names it
Both QuickML endpoints unavailable	Bundled backup model answers, labelled on screen with a warning
Application unreachable during evaluation	predict.py retries, then scores with a built-in linear model (9.3 kW RMSE)
Malformed CSV input	Columns matched, blanks filled, ranges clipped, timestamps interpolated
Invalid dashboard input	Rejected with a specific message, e.g. temperature outside –60 to 60 °C
Large batch	Sent in 2,000-row chunks; 10,000 rows scored in 21 seconds
Excess request volume	Per-client rate limits and request-size limits
Data Store failure	Persistence runs after the response; forecasts are unaffected
Credential exposure	No keys in the browser, the repository or predict.py

Verification results

Check	Result
Automated test suite	891 of 891 passed
Organisers’ self-test, python test_eval.py, clean environment without credentials	SUCCESS · 1,000 rows in 2.3 s
Batch scoring through the deployed API	10,000 rows in 21 s
Application predictions against direct QuickML endpoint calls	Identical for the five hours compared
Data Store persistence for one dashboard forecast	1 run, 24 telemetry rows, 6 alert events
Reopening a saved run	Restored from Data Store without a model call
Dashboard at desktop and 390 px mobile widths	All views rendered; no console errors
Exports	Predictions CSV, schedule CSV, dispatch JSON

Checks run on the live Development deployment, 24–25 October 2026

Operator dashboard



View	Content
Overview	Peak, energy, storage hours, overloads; demand chart with thresholds
Dispatch	24-hour schedule and transition controller log
Battery	State-of-charge chart and gauge
Scenario lab	Six-capacity comparison and sizing
Inputs	CSV upload, hourly editor, manual form
System log	Saved runs and events from Data Store

Screenshot of the live Development deployment

Deployment and access

Item	Details
Dashboard	`gridwise-50047143069.development.catalystappsail.in`
Evaluation command	`python predict.py --input test.csv --output predictions.csv`
Model	QuickML endpoints gridwise-demand-catboost and gridwise-demand-linear · IN data centre
Data Store tables	ForecastRuns · Telemetry · AlertLog
Source repository	`github.com/Avi36005/gridwise-hacksphere-3`

Thank you

Team Kryptonite · Gridwise